R_PROGRAMMING_LAB_OUTPUTS(11249A370)

EX1.01:

```
> roll_no
[1] 101
> name
[1] "janani"
> cgpa
[1] 8.93
> pass
[1] TRUE


>
```

EX1.02:

```
[1] 5001
> salary
[1] 45000.75
> department
[1] "CSE"
> is_permanent
[1] TRUE


>
```

EX1.03:

```
> product_name
[1] "mobile"
> price
[1] 25000
> stock_available
[1] 25
> discount_eligible
[1] FALSE
```

EX1.04:

```
> city
[1] "KANCHIPURAM"
> temperature
[1] 36.5
> rain
[1] FALSE
> wind_speed
[1] 12.8
```

EX1.05:

```
> subject
[1] "Mathematics"
```

```
> total_marks
[1] 100
> marks_obtained
[1] 78
> result
[1] TRUE
```

EX1.06:

```
> account_no
[1] 123456789
> holder_name
[1] "Ravi chandra"
> balance
[1] 25000.5
> account_active
[1] TRUE
```

EX1.07:

```
> movie_title
[1] "Inception"
> rating
[1] 7.9
> reviews
[1] 1200
> available
[1] TRUE
```

EX1.08:

```
> vehicle_count
[1] 350
> location
[1] "BYPASS Road"
> avg_speed
[1] 45.6
> congested
[1] TRUE
```

EX2.01:
[1] "Eligible for Loan"

EX2.02:
[1] "Eligible for Exam"
EX2.03:
[1] "Discount Applied"
EX2.04:
[1] "High Usage"
EX2.05:
[1] "Bonus Eligible"
EX2.06:
[1] "Valid Password"

EX2.07:
[1] "Extreme Temperature Warning"
EX2.08:
[1] "A"
EX2.09:
[1] "Credit Card Approved"

EX3.01:
[1] 800
EX3.02:
[1] "Pass"
EX3.03:
> net_salary(30000)
[1] 27000
EX3.04:
> electricity_bill(250)
[1] 750
EX3.05:
> final_amount(4500)
[1] 4050
EX3.06:
> celsius_to_fahrenheit(37)
[1] 98.6
EX3.07:
> assign_grade(91)
[1] "A"
EX3.08:
> check_voting(20)
[1] "Eligible to Vote"

EX3.09:

> calculate_emi(500000, 7.5, 10)
[1] 5935.09

EX3.10:

> hospital_bill(500, 2000, 3)
[1] 6825

EX3.11:
> order_total(4200)
[1] 3880
EX3.12:
> calculate_gpa(c(78, 85, 90, 72, 80))
[1] "First Class"
> [1] 3880
EX3.13:

> smart_meter_bill(180, 150)
[1] 1683

EX3.14:

```
> insurance_premium(5000, 55, TRUE)
[1] 7800


>
```

EX3.15:

```
> payroll(40000)
[1] 45760
```

EX3.16:

```
> water_usage_status(1200)
[1] "Critical"
```

EX4.01:

```
RollNo    Name Dept Marks
1    101    Ravi  CSE    78
2    102 Chandra  ECE    85
3    103 Karthik   IT    90
4    104  Anjali  CSE    88
```

EX4.02:

```
> employees
  EmpID   Name    Dept Salary
1     1   Amit      HR  35000
2     2   siva      IT  45000
3     3 sakthi Finance  50000
4     4   Neha      IT  48000
```

EX4.03:

```
> patient_records
  PatientID    Name Age  Cost
1       201  Suresh  45 12000
2       202 Lakshmi  38 15000
```

EX4.05:

```
> patient_records
  PatientID    Name Age  Cost
1       201  Suresh  45 12000
2       202 Lakshmi  38 15000
```

EX4.06:

```
> payroll
  EmpID  Name    Dept Basic  HRA   DA Rating
1   101  Rahul     IT  30000  6000 3000    4.5
2   102  Sneha     HR  35000  7000  3500    4.2
```

```
3   103  Kiran  Finance  40000  8000  4000    4.8
```

EX4.07:

```
> hospital_data
  PatientID    Name Age      Disease Amount
1       201  Suresh  45     Diabetes  15000
2       202 Lakshmi  38 Hypertension  12000
3       203   Arjun  52      Cardiac  25000
```

EX4.08:

```
> sales_data
  ProductID  ProductName  UnitsSold Revenue
1         1      Mobile       120 1200000
2         2      Laptop        80 4000000
3         3      Tablet        60 1800000
4         4 Smart Watch        90  900000
```

EX4.09:

```
> agri_data
  FieldID  Crop Moisture Temp ExpectedYield
1       1  Rice       45   32          4.5
2       2 Wheat       38   29          3.8
3       3 Maize       40   31          4.1
```

EX4.10:

```
> final_records
  RollNo    Name Dept Sem1 Sem2 Attendance
1      1    Arun  CSE   78   80         92
2      2 SRIVANI  ECE   82   85         88
3      3  Charan   IT   90   88         95
```

EX4.11:

```
> payroll
  EmpID  Name    Dept Basic  HRA   DA Rating
1   101 Rahul      IT 30000 6000 3000    4.5
2   102 Sneha      HR 35000 7000 3500     NA
3   103 Kiran Finance    NA   NA   NA    4.8
```

EX4.12:

```
> hospital_data
  PatientID   Name Age  Disease Amount
1       201 Suresh  45 Diabetes  15000
2       202   jaya  38     <NA>  12000
3       203  Arjun  52  Cardiac  25000
```

EX4.13:

```
> sales_data
  ProductID ProductName UnitsSold Revenue
1         1      Mobile       120 1200000
```

```
2        2     Laptop       80 4000000
3        3     Tablet       60      NA
4        4 Smart Watch       NA  900000
```

EX4.14:

```
> agri_data
  FieldID  Crop Moisture Temp ExpectedYield
1       1  Rice       45   32          4.5
2       2 Wheat       38   29           NA
3       3 Maize       NA   NA          4.1
```

EX5.01:

```
> upper_names
[1] "ARUN"   "BALA"   "CHITRA"
> name_length
[1] 4 4 6
```

EX5.02:

```
> usernames
[1] "arun"  "meena" "ravi"
```

EX5.03:

```
> clean_input
[1] "R Programming" "Data Science"  "AI"
```

EX5.04:

```
> valid_codes
[1]  TRUE  TRUE FALSE
```

EX5.05:

```
> updated_report
[1] "issue in file, issue in system, issue occurred"
```

EX5.06:

```
[1] "Strong Password"
```

EX5.07:

```
> word_count
[1] 6
```

EX5.08:

```
> extensions
[1] "csv" "pdf" "png"
```

EX5.09:

```
> abbr
[1] "sri" "Kar" "Ani"
```

EX5.10:

```
> matched_names
[1] "Bala" "Ravi" "jai"
```

EX5.11:

```
> clean_students
     Name         Email Department Username ValidEmail
1 GAYATHRI gayu@univ.edu        CSE     gayu       TRUE
2     RAVI ravi@univ.edu        ECE     ravi       TRUE
```

EX5.12:

```
> products
     Product       Code ProductCode
1 Apple Phone PRD-101-APL        APL
2  Samsung-Tv PRD-202-SAM        SAM
3 Dell Laptop PRD-303-DEL        DEL
```

EX5.13:

```
> comments
  User           Comment                CleanComment
1    A     this app is bad     this app is unacceptable
2    B very bad experience very unacceptable experience
3    C      not bad at all      not unacceptable at all
```

EX6.01:

```
> highest <- max(marks)
> total
[1] 417
> average
[1] 83.4
> highest
[1] 90
```

EX6.02:

```
> min_temp
[1] 31
>
> max_temp
[1] 36
```

EX6.03:

```
> employee$Name
[1] "Gayathri"
> employee$Salary
[1] 45000
```

EX6.04:

```
$TestResults
[1] 120 140 130
> patient$TestResults
[1] 120 140 130
```

EX6.05:

```
> students
  RollNo    Name Dept Marks
1      1  Anitha  CSE    85
2      2 Karthik  ECE    78
3      3   Meena   IT    92
```

EX6.06:

```
> total_sales
[1] 170000
```

EX6.07:

```
> orders
    Item Price
1   Phone  1200
2 Headset   850
3 Charger   560
```

EX6.08:

```
> class_results
  RollNo Total
1    101   240
2    102   255
3    103   270
```

EX6.09:

```
> transactions
   Date Amount
1 01-01   -500
2 02-01   1000
3 03-01   -200
```

EX6.10:

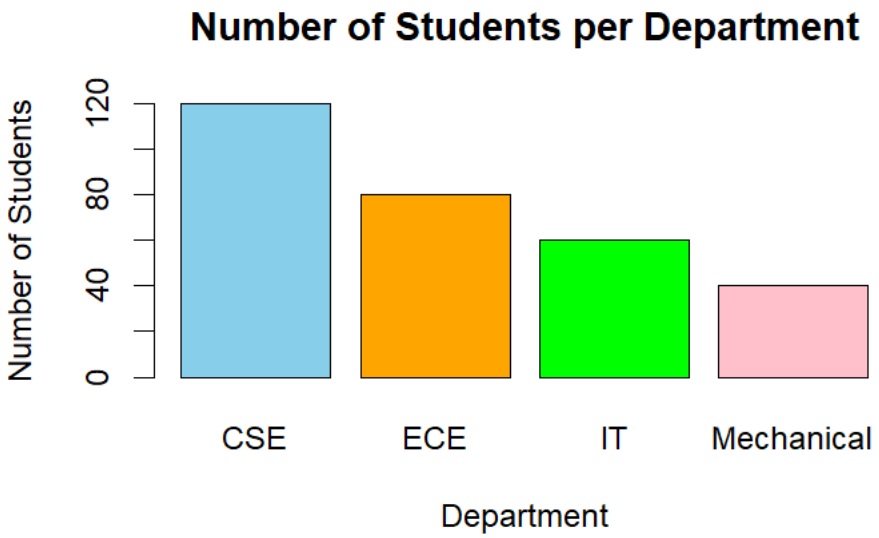
```
> profile
$Name
[1] "Gayathri"

$Age
[1] 21

$Scores
[1] 85 90 88

> records
  Subject Marks
1    Math    85
2      CS    90
3 Physics    88
```

EX7.01:

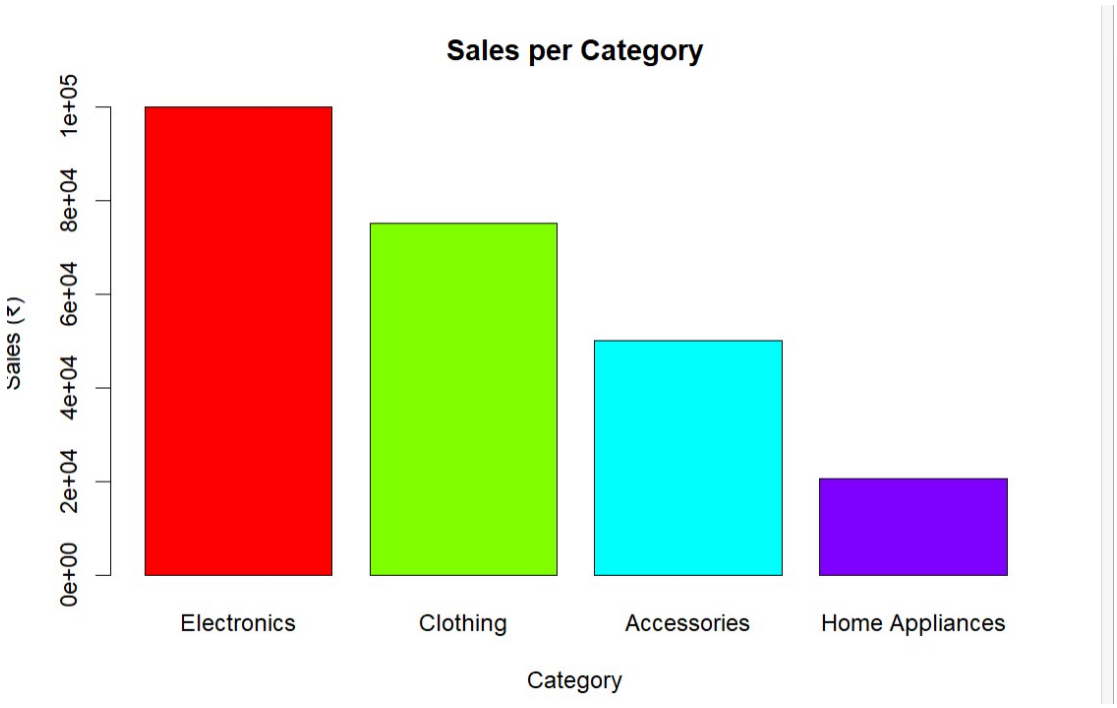
```
# Load CSV file
> sales <- read.csv("sales_data.csv")
>
> # View data
> head(sales)
       Date Product    Category Quantity Price
1 2026-01-01   Phone Electronics        5 20000
2 2026-01-01     Bag Accessories       10  1500
3 2026-01-02  Laptop Electronics        3 50000
4 2026-01-02 Headset Electronics        7  2500
5 2026-01-03     Bag Accessories        5  1500
6 2026-01-03   Phone Electronics        3 20000
> str(sales)
```

```
'data.frame':	6 obs. of  5 variables:
 $ Date    : chr  "2026-01-01" "2026-01-01" "2026-01-02" "2026-01-02" ...
 $ Product : chr  "Phone" "Bag" "Laptop" "Headset" ...
 $ Category: chr  "Electronics" "Accessories" "Electronics" "Electronics" ...
 $ Quantity: int  5 10 3 7 5 3
 $ Price   : int  20000 1500 50000 2500 1500 20000
> summary(sales)
    Date           Product           Category           Quantity
 Length:6           Length:6           Length:6           Min.   : 3.0
 Class :character   Class :character   Class :character   1st Qu.: 3.5
 Mode  :character   Mode  :character   Mode  :character   Median : 5.0
                                                          Mean   : 5.5
                                                          3rd Qu.: 6.5
                                                          Max.   :10.0
    Price
 Min.   : 1500
 1st Qu.: 1750
 Median :11250
 Mean   :15917
 3rd Qu.:20000
 Max.   :50000
>
> # Data Cleaning
> sales$Product <- trimws(sales$Product)
> sales$Category <- trimws(sales$Category)
> sales$Date <- as.Date(sales$Date, format="%Y-%m-%d")
>
> # Add TotalSales column
> sales$TotalSales <- sales$Quantity * sales$Price
>
> # Total sales per product
> total_sales_product <- aggregate(TotalSales ~ Product, data=sales, sum)
> print(total_sales_product)
  Product TotalSales
1     Bag      22500
2 Headset      17500
3  Laptop     150000
4   Phone     160000
>
> # Top-selling product
> top_product <- total_sales_product[which.max(total_sales_product$TotalSales), ]
> print(top_product)
  Product TotalSales
4   Phone     160000
>
> # Total quantity per category
> total_qty_category <- aggregate(Quantity ~ Category, data=sales, sum)
> print(total_qty_category)
    Category Quantity
1 Accessories       15
2 Electronics       18
```

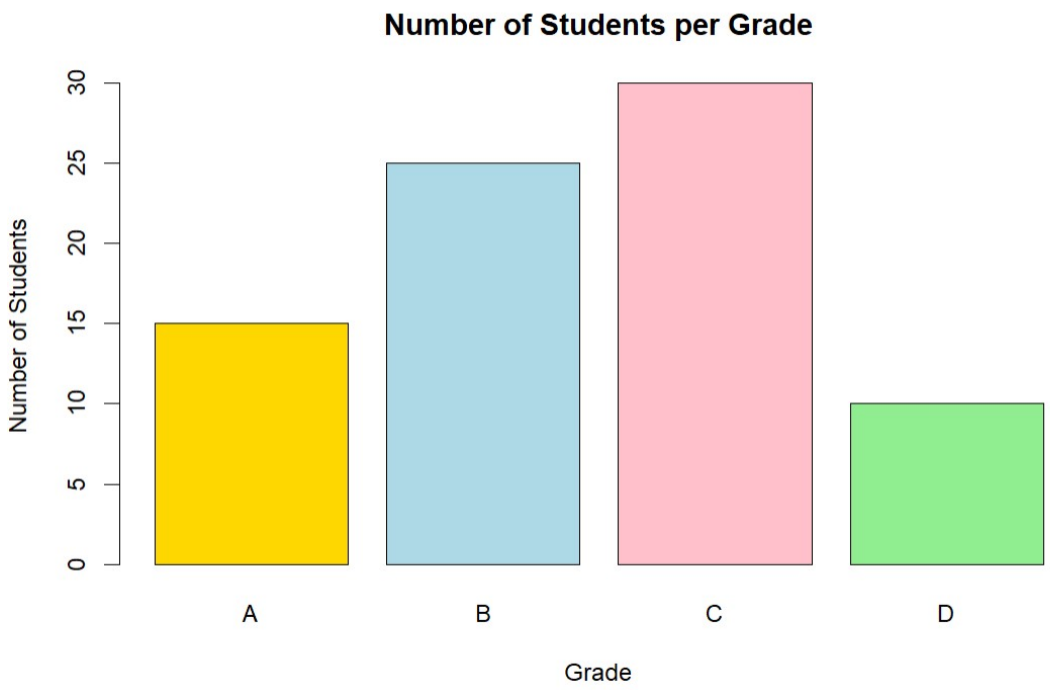
EX8.01:


Number of Students per Department
Number of Students
0
40
80
120
CSE
ECE
IT
Mechanical
Department

EX8.02:


Sales per Category
Sales (₹)
0e+00
2e+04
4e+04
6e+04
8e+04
1e+05
Electronics
Clothing
Accessories
Home Appliances
Category

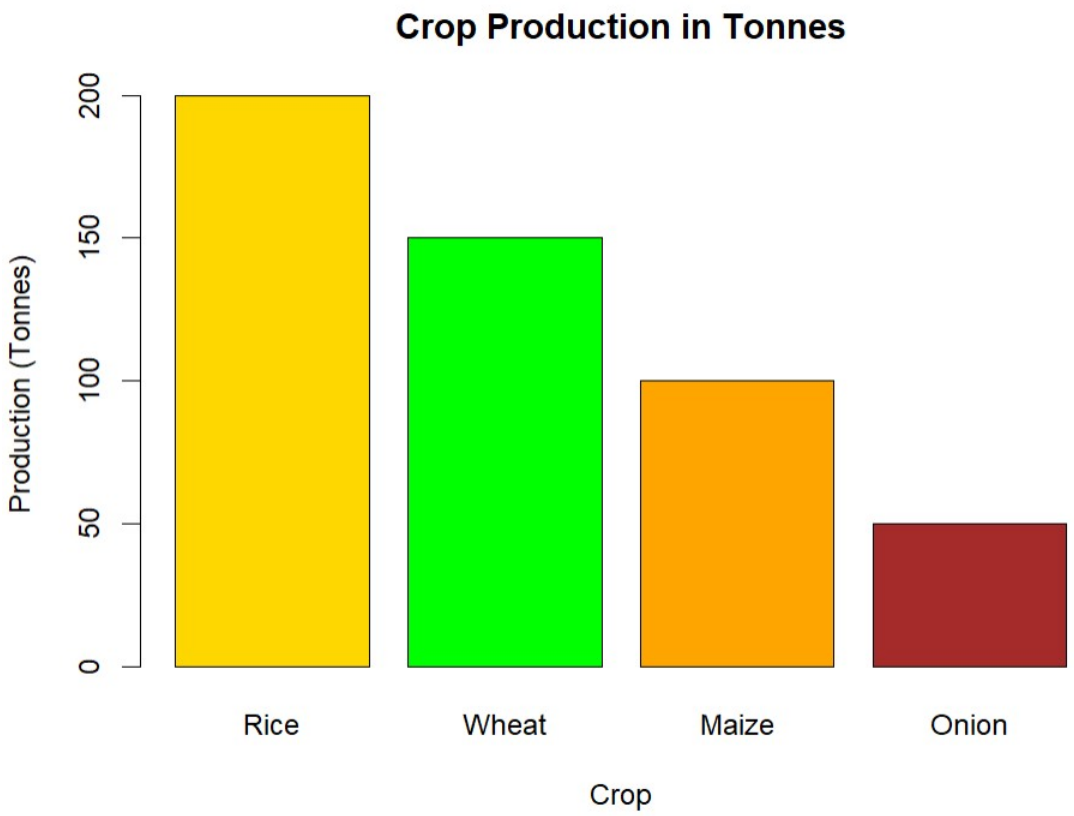
EX8.03:


Number of Students per Grade
Number of Students
0
5
10
15
20
25
30
A
B
C
D
Grade

EX8.04:

Number of Employees by Experience
Number of Employees
0
20
40
60
80
0-2 years
3-5 years
6-10 years
10+ years
Experience Level

EX8.05:



EX9.01:

```
> mean_marks  Maths Science English
   82.9    80.0    85.4 > median_marks  Maths
Science English
   85.5    81.0    86.0 > sd_marks   Maths
Science  English
9.302927 9.809293 7.183314 > top_student
Name Maths Science English Total
4 kamala    95      92      98   285>
above_avg_math        Name Maths Science
English Total
1        rama    85      78      92   255
2   Meenakshi    90      88      80   258
4      kamala    95      92      98   285
6       Sneha    88      80      85   253
8    Gayathri    92      90      91   273
10    Lakshmi    86      82      87   255


>
```

EX9.02:

```
> sales_stats   Product Mean Max Min Variance
1     Phone   5.4   7   4      1.3
2    Laptop   3.4   5   2      1.3
3   Headset   6.6   8   5      1.3
4       Bag  10.0  12   8      2.5
5    Camera   2.2   3   1      0.7
6     Mouse   7.6   9   6      1.3
7 Keyboard   6.4   8   5      1.3>
max_avg_product  Product Mean Max Min Variance
4      Bag   10  12   8      2.5
```

```
>
```

EX9.03:

```
> mean_age[1] 45.4
> median_age[1] 46.5
> range_age[1] 29 60
> sd_age[1] 9.547542
> older_patients
   PatientID  Name Age
3        203 Arjun  52
5        205  Ravi  60
7        207 Kiran  55

>
```

EX9.04:

```
> dept_stats
    Dept Salary.Mean Salary.Median
1 Finance    42333.33         42000
2      HR    36000.00         36000
3      IT    31500.00         31500
> high_earners
EmpID      Name    Dept Salary
3   103     Kiran Finance  40000
6   106 Gayathri Finance  42000
8   108  Lakshmi      HR  37000
9   109    Arjun Finance  45000

>
```

EX9.05:

```
> mean_rating
[1] 4.1
> median_rating
[1] 4
> mode_rating
[1] 4
> var_rating
[1] 0.5444444
> top_rated
ProductID Product Rating
2         2  Laptop      5
5         5  Camera      5
8         8 Charger      5

>
```